REINFORCEMENT LEARNING WITH VERIFIABLE REWARDS IMPLICITLY INCENTIVIZES CORRECT REASONING IN BASE LLMS

Xumeng Wen[*1], **Zihan Liu**[*†2], **Shun Zheng**[*‡1], **Shengyu Ye**[†1], **Zhirong Wu**[1], **Yang Wang**[1], **Zhijian Xu**[†3], **Xiao Liang**[†4], **Junjie Li**[1], **Ziming Miao**[1], **Jiang Bian**[1], **Mao Yang**[1]
[1]Microsoft Research Asia [2]Peking University
[3]The Chinese University of Hong Kong [4]University of California, Los Angeles

arXiv:2506.14245v2 [cs.AI] 2 Oct 2025

ABSTRACT

Recent advancements in long chain-of-thought (CoT) reasoning, particularly through the Group Relative Policy Optimization algorithm used by DeepSeek-R1, have led to significant interest in the potential of Reinforcement Learning with Verifiable Rewards (RLVR) for Large Language Models (LLMs). While RLVR promises to improve reasoning by allowing models to learn from free exploration, there remains debate over whether it truly enhances reasoning abilities or simply boosts sampling efficiency. This paper systematically investigates the impact of RLVR on LLM reasoning. We revisit Pass@K experiments and demonstrate that RLVR can extend the reasoning boundary for both mathematical and coding tasks. This is supported by our introduction of a novel evaluation metric, CoT-Pass@K, which captures reasoning success by accounting for both the final answer and intermediate reasoning steps. Furthermore, we present a theoretical framework explaining RLVR's incentive mechanism, demonstrating how it can encourage correct reasoning even when rewards are based solely on answer correctness. Our analysis of RLVR's training dynamics reveals that it incentivizes correct reasoning early in the process, with substantial improvements in reasoning quality confirmed through extensive evaluations. These findings provide strong evidence of RLVR's potential to enhance LLM reasoning, offering valuable insights into its mechanisms and performance improvements.

1 INTRODUCTION

The successful replication of long chain-of-thought (CoT) reasoning, similar to that in OpenAI's o1 (OpenAI, 2024), by DeepSeek-R1 (Guo et al., 2025) using the Group Relative Policy Optimization (GRPO) algorithm (Shao et al., 2024), has sparked a surge of interest within the open research community. This interest is focused on understanding, reproducing, and extending DeepSeek's approach, as evidenced by a multitude of recent studies (Liu et al., 2025b; Hu et al., 2025; Zeng et al., 2025; Yu et al., 2025; He et al., 2025; Wen et al., 2025; Chen et al., 2025b). Fundamentally, this emerging paradigm is a form of Reinforcement Learning with Verifiable Rewards (RLVR) (Lambert et al., 2024; Guo et al., 2025; Yue et al., 2025), where a Large Language Model (LLM) acts as a policy, generating a CoT as a sequence of actions and receiving feedback on answer correctness from deterministic verifiers. This paradigm holds the promise of endowing LLMs with the ability to learn from experience through free exploration, potentially leading to unlimited intelligence (OpenAI, 2024; Guo et al., 2025; Silver & Sutton, 2025).

However, emerging concerns question the true effectiveness of RLVR. These concerns are motivated by the observation that some post-RLVR models improve the Pass@1 metric but fail to enhance the Pass@K metric compared to the base (pre-RLVR) model. This phenomenon was first noted by Shao et al. (2024) during the development of GRPO. Subsequently, a systematic study by Yue et al. (2025)

[*]These authors contributed equally: Xumeng Wen, Zihan Liu, Shun Zheng.
[†]Work done during the internship at Microsoft Research Asia.
[‡]Correspondence to shun.zheng@microsoft.com.

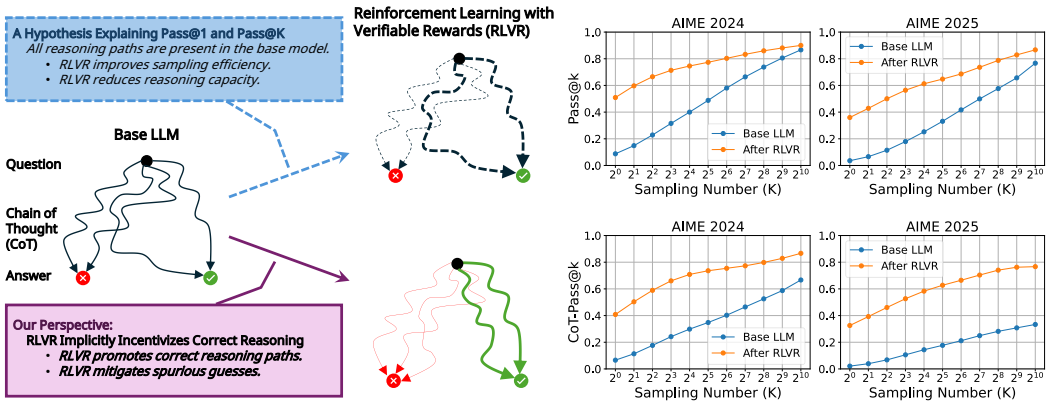

Figure 1: An illustration of our perspective: **RLVR implicitly incentivizes correct reasoning in base LLMs**. We visualize how different explanation frameworks lead to varying reasoning paths being activated, with our perspective shown in the lower left and a recent popular hypothesis explaining Pass@K observations (Yue et al., 2025) summarized in the upper left. In this diagram, the line width represents the sampling probability of a reasoning path, while the color distinguishes correct paths (green) from incorrect ones (red). If all reasoning paths after applying RLVR are already present in the base model, the reasoning model merely adjusts the sampling probabilities of these existing paths (visualized in dashed lines). This hypothesis effectively accounts for the key observation shown in the upper-right part, where, for a moderately large K, a base LLM can catch up to the reasoning model after RLVR using the Pass@K metric. In this study, we unveil **the extended reasoning capability boundary** in math tasks using a refined metric, **CoT-Pass@K**, which emphasizes both the correctness of answers and the validity of reasoning CoTs.

on more open-weight RLVR models discovered that the Pass@K metric of the base model increases at a much faster rate than its RLVR-tuned counterpart. Consequently, for a moderately large K, the base model eventually matches and surpasses the reasoning model. This led to their adventurous hypothesis: all correct reasoning paths are already present in the base model, and RLVR merely improves sampling efficiency at the cost of reducing overall reasoning capacity.

While this hypothesis has gained significant support (Zhu et al., 2025; Zhang et al., 2025; Wang et al., 2025a; Chen et al., 2025a), conflicting observations have also been reported. For instance, Liu et al. (2025a) detected the emergence of new reasoning patterns after RLVR, while they also acknowledged a loss in reasoning capacity as measured by Pass@K. Chen et al. (2025b) reported persistent improvements in Pass@K for competitive coding tasks but did not show improved Pass@K for math tasks. Shojaee et al. (2025) observed similar Pass@K observations on math datasets but found different patterns on puzzles with high complexity. To the best of our knowledge, no systematic explanation exists to reconcile these contradictory findings, leaving a critical question unanswered: "*should we accept the hypothesis as a fundamental limitation of RLVR or should we trust new empirical findings that challenge the hypothesis?*"

In this work, we address this debate systematically and demonstrate that RLVR can fundamentally enhance the reasoning abilities of LLMs. First, we revisit Pass@K experiments and unveil the existence of extended reasoning capability boundaries after RLVR for both math and code tasks. In addition to reproducing the extended reasoning boundary in competitive coding, as reported by Chen et al. (2025b), we find that the Pass@K performance of base LLMs on math reasoning can be unreliable, as base LLMs are capable of producing incorrect CoTs yet coincidentally arriving at the ground truth, especially for hard mathematical questions where answers are simple and can be easily guessed after multiple attempts. To address this, we introduce a new metric, CoT-Pass@K, which evaluates success only when both the final answer and the intermediate reasoning CoT are correct. In practice, we verify the correctness of mathematical CoTs by instructing DeepSeek-R1-0528-Qwen3-8B (DeepSeek, 2025) and confirm their reliability. Using this new metric, we successfully identify the extended reasoning boundary of a post-RLVR model for math tasks. Figure 1 summarizes our key perspectives.

Moreover, we develop a theoretical framework to explain why RLVR works, even when base LLMs may guess the ground truth and only answer correctness is provided as a reward, and how RLVR incentivizes correct reasoning. Our central insight is that once LLMs have been pre-trained to establish strong knowledge and logic priors that distinguish correct from incorrect CoTs, the GRPO gradient will increase the probability of generating more correct CoTs.

Additionally, we investigate the training dynamics of RLVR to understand when this improved reasoning emerges. By reproducing GRPO-style training using the open-source DAPO recipe (Yu et al., 2025) and performing extensive verifications, we find that RLVR begins to incentivize correct reasoning from the early stages of training, and this capability generalizes well to unseen test questions. The results of our training analysis align with our theorem, which highlights the implicit incentivization of correct reasoning CoTs.

Finally, we evaluate the quality of generated CoTs from a learning perspective: if supervised learning on some CoT data results in better generalization performance on test sets, we regard them as high quality. This allows us to evaluate the quality of CoTs generated by model checkpoints at different RLVR stages. Our results show that after RLVR, the quality of reasoning CoTs has been fundamentally improved.

In summary, our contributions include:

- A systematic evaluation revealing the extended reasoning capability boundary after RLVR for both code and math tasks.
- A theoretical understanding of why RLVR works with only answer correctness as a reward and how RLVR incentivizes correct reasoning.
- An analysis of RLVR's training dynamics, delving deeper into optimization effects, generalization behaviors, and current limitations.
- Confirmation of the quality improvements in reasoning CoTs from a learning perspective, replicating the generalization abilities of post-RLVR models trained with enormous costs simply via supervised fine-tuning.

2 Related Work

RLVR Since the release of DeepSeek-R1 (Guo et al., 2025), there has been a surge of research interest in the RLVR paradigm (Luo et al., 2025b; Liu et al., 2025b; Hu et al., 2025; Cui et al., 2025; Xie et al., 2025; Zeng et al., 2025; Yu et al., 2025; Luo et al., 2025a; Chen et al., 2025a; He et al., 2025; Wen et al., 2025; Cao et al., 2025; Liu et al., 2025a; Chen et al., 2025b). Due to the high computational cost of RLVR, most studies have focused on small- to medium-sized models (up to 32B parameters). These studies span a wide range of aspects, including training data curation, objective design, hyperparameter tuning, base model selection, and various insightful observations. However, only a few studies have addressed the theoretical foundations of RLVR. In this work, we argue that RLVR for LLMs should be understood from a different perspective—one that emphasizes the correctness of reasoning paths. We hope our empirical findings and theoretical perspective could inspire the community to develop more efficient and effective RLVR approaches, unlocking its broader potential across diverse applications.

Debates on Whether RLVR Really Incentivizes Since Yue et al. (2025) raised the insightful question of whether RLVR truly incentivizes improvements beyond the base LLMs, and conducted extensive empirical experiments to demonstrate the wide applicability of their key hypothesis—that RLVR does not improve $Pass@K$ for the base LLM because all reasoning paths are already present in the base model—there have been varying perspectives on this hypothesis. Some studies agree with this viewpoint (Wang et al., 2025b; Zhu et al., 2025; Zhang et al., 2025; Wang et al., 2025a; Chen et al., 2025a), while others report contradictory findings (Liu et al., 2025a; Chen et al., 2025b; Shojaee et al., 2025), as discussed in the introduction. There is currently no fundamental understanding to resolve these debates. Liu et al. (2025a) speculated that previous RLVR experiments may have been conducted within a single domain (e.g., math) and were optimized for limited gradient steps before true exploration could occur. Shojaee et al. (2025) suggested that the complexity of puzzles might be the key factor. Chen et al. (2025b) presented statistically significant empirical results to justify that their model indeed improves $Pass@K$, particularly highlighting a persistent